

Finite Blocklength MIMO Precoding With Mixed Power and QoS Constraints

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Paper Explanation Presentation

Based on: Jingchen Wu et al., *IEEE Communications Letters*, 2026

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Layer 0: Overall Idea of the Paper

Core Problem

Design downlink MIMO precoders that maximize finite-blocklength weighted sum rate while satisfying mixed power constraints and QoS requirements.

- ▶ To reduce difficulty of precoder design problem, it is reformulated into an easier SINR allocation problem.

Hard precoder design $\implies$ SINR allocation problem

- ▶ Instead of directly optimizing the precoding matrix V , the paper optimizes target SINRs γ_k .
- ▶ A power feasibility function $P(\gamma)$ checks whether the target SINRs can be achieved.
- ▶ Two algorithms are proposed:
 - ▶ BDGM: bi-direction gradient method.
 - ▶ IWF: iterative water-filling algorithm.

Layer 1: Main Research Gap

- ▶ Existing finite-blocklength precoding works usually focus on limited settings.
- ▶ Many methods consider only a single sum power constraint:

$$\sum_{k=1}^K \|v_k\|^2 \leq P$$

- ▶ Practical MIMO systems may have different power limits:
 - ▶ sum power constraint,
 - ▶ per-antenna power constraint,
 - ▶ per-group or per-AP power constraint.
- ▶ Therefore, a general finite-blocklength precoding framework is needed for mixed power constraints and QoS guarantees.

Layer 2: Downlink MIMO System Model

- ▶ The system has N transmit antennas serving K single-antenna users.
- ▶ The transmitted signal is:

$$\mathbf{x} = \sum_{k=1}^K \mathbf{D}_k \mathbf{v}_k s_k$$

- ▶ Variables:
 - ▶ $\mathbf{x} \in \mathbb{C}^N$: transmitted signal vector from all antennas.
 - ▶ N : number of transmit antennas.
 - ▶ K : number of users.
 - ▶ k : user index, where $k = 1, \dots, K$.
 - ▶ s_k : data symbol intended for user k .
 - ▶ $\mathbb{E}[s_k] = 0$, $\mathbb{E}[|s_k|^2] = 1$: symbol has zero mean and unit power.
 - ▶ $\mathbf{v}_k \in \mathbb{C}^N$: precoding vector for user k .
- ▶ Antenna-user association:

$$\mathbf{D} = [\mathbf{d}_1, \dots, \mathbf{d}_K] \in \{0, 1\}^{N \times K}$$

$$[\mathbf{D}]_{nk} = \begin{cases} 1, & \text{if antenna } n \text{ serves user } k, \\ 0, & \text{otherwise.} \end{cases}$$

Layer 2: Received Signal Model

$$y_k = \mathbf{h}_k^H \mathbf{D}_k \mathbf{v}_k s_k + \sum_{j \neq k}^K \mathbf{h}_k^H \mathbf{D}_j \mathbf{v}_j s_j + n_k$$

- ▶ y_k : signal received by user k .
- ▶ $\mathbf{h}_k \in \mathbb{C}^N$: channel vector from all N antennas to user k .
- ▶ $(\cdot)^H$: Hermitian transpose.
- ▶ $\mathbf{h}_k^H \mathbf{D}_k \mathbf{v}_k s_k$: desired signal for user k .
- ▶ $\sum_{j \neq k}^K \mathbf{h}_k^H \mathbf{D}_j \mathbf{v}_j s_j$: interference from other users.
- ▶ j : index of another user, where $j \neq k$.
- ▶ n_k : additive noise at user k .

Layer 2: SINR of Each User

$$\rho_k(\mathbf{V}) = \frac{|\mathbf{h}_k^H \mathbf{D}_k \mathbf{v}_k|^2}{\sum_{\substack{j=1 \\ j \neq k}}^K |\mathbf{h}_k^H \mathbf{D}_j \mathbf{v}_j|^2 + \sigma_k^2}$$

- ▶ $\rho_k(\mathbf{V})$: SINR of user k .
- ▶ $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_K] \in \mathbb{C}^{N \times K}$: complete precoding matrix.
- ▶ Numerator:

$$|\mathbf{h}_k^H \mathbf{D}_k \mathbf{v}_k|^2$$

is the useful received signal power for user k .

- ▶ Denominator:

$$\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{D}_j \mathbf{v}_j|^2 + \sigma_k^2$$

is interference power plus noise power.

- ▶ σ_k^2 : noise power at user k . The paper assumes $\sigma_k^2 = 1$ for simplicity.

Layer 3: Finite Blocklength Rate Model

$$R_k(\rho_k) = \log(1 + \rho_k) - \frac{Q^{-1}(\epsilon_k)}{\sqrt{M}} \sqrt{B(\rho_k)}$$

$$B(\rho_k) = 1 - (1 + \rho_k)^{-2}$$

- ▶ $R_k(\rho_k)$: achievable finite-blocklength rate of user k .
- ▶ ρ_k : SINR of user k .
- ▶ $\log(1 + \rho_k)$: classical infinite-blocklength Shannon rate term.
- ▶ M : blocklength, i.e., number of channel uses used to transmit a codeword.
- ▶ ϵ_k : target block error rate of user k .
- ▶ $Q^{-1}(\epsilon_k)$: inverse Gaussian Q -function; represents reliability penalty.
- ▶ $B(\rho_k)$: channel dispersion term.
- ▶ The second term reduces the rate because short-packet transmission cannot achieve Shannon capacity.

Layer 4: Mixed Power Constraints

$$\sum_{k=1}^K \mathbf{v}_k^H \mathbf{S}_i \mathbf{v}_k \leq P_i, \quad \forall i = 1, \dots, L$$

- ▶ i : index of a power constraint.
- ▶ L : total number of power constraints.
- ▶ $\mathbf{v}_k$: precoder for user k .
- ▶ $\mathbf{S}_i \in \mathbb{C}^{N \times N}$: positive semi-definite matrix selecting which antenna, group, or total power is constrained.
- ▶ P_i : power budget of constraint i .
- ▶ $\mathbf{v}_k^H \mathbf{S}_i \mathbf{v}_k$: contribution of user k 's precoder to power constraint i .

Layer 4: Types of Power Constraints

- ▶ The mixed power constraint framework can represent different practical power limits.

- ▶ **Sum Power Constraint (SPC):**

$$\mathbf{S}_i = \mathbf{I}_N$$

This limits the total transmit power across all antennas.

- ▶ **Per-Antenna Power Constraint (PAPC):**

$$\mathbf{S}_i = \mathbf{e}_{m_i} \mathbf{e}_{m_i}^H$$

This limits the transmit power of antenna m_i .

- ▶ **Per-Group Power Constraint (PGPC):**

$$\mathbf{S}_i = \sum_{m \in \mathcal{G}_i} \mathbf{e}_m \mathbf{e}_m^H$$

This limits the total power of antenna group $\mathcal{G}_i$.

- ▶ m : antenna index, where $m = 1, \dots, N$, $[\mathbf{v}_k]_{m_i}$: the precoder coefficient applied at antenna m_i for user k . , m_i : the specific antenna index selected by constraint i , $\mathbf{e}_{m_i}$: unit vector that selects antenna m_i .. $\mathcal{G}_i$: set of antenna indices belonging to group i .

Layer 4: How $\mathbf{S}_i$ Creates Different Power Constraints

$$\sum_{k=1}^K \mathbf{v}_k^H \mathbf{S}_i \mathbf{v}_k \leq P_i$$

- ▶ $\mathbf{S}_i$ acts like a power selection matrix.
- ▶ If $\mathbf{S}_i = \mathbf{I}_N$:

$$\sum_{k=1}^K \mathbf{v}_k^H \mathbf{v}_k \leq P_i$$

This becomes a total sum power constraint.

- ▶ If $\mathbf{S}_i = \mathbf{e}_{m_i} \mathbf{e}_{m_i}^H$:

$$\sum_{k=1}^K |[\mathbf{v}_k]_{m_i}|^2 \leq P_i$$

This limits only antenna m_i .

- ▶ If $\mathbf{S}_i = \sum_{m \in \mathcal{G}_i} \mathbf{e}_m \mathbf{e}_m^H$:

$$\sum_{k=1}^K \sum_{m \in \mathcal{G}_i} |[\mathbf{v}_k]_m|^2 \leq P_i$$

This limits the total power of antenna group $\mathcal{G}_i$.

- ▶ Thus, changing $\mathbf{S}_i$ changes what part of the precoder power is measured.

Layer 4: Normalized Mixed Power Constraint

$$\mathbf{X}_i = \frac{\mathbf{S}_i}{P_i}$$

$$\sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \leq 1, \quad \forall i = 1, \dots, L$$

Since we want this power constraint for every constraint $i \in \{1, \dots, L\}$

$$\max_{1 \leq i \leq L} \left\{ \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \right\} \leq 1$$

- ▶ $\mathbf{X}_i$: normalized power selection matrix for constraint i .
- ▶ $\mathbf{S}_i$: original power selection matrix.
- ▶ P_i : power budget of constraint i .
- ▶ Normalization converts every power constraint into the same upper bound 1.
- ▶ The maximum form means all power constraints must be satisfied simultaneously.

Layer 4: Why the Maximum Form is Used

$$\sum_{k=1}^K \mathbf{v}_k^H \mathbf{S}_i \mathbf{v}_k \leq P_i, \quad \forall i = 1, \dots, L$$

$$\mathbf{X}_i = \frac{\mathbf{S}_i}{P_i}$$

$$\sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \leq 1, \quad \forall i = 1, \dots, L$$

$$\max_{1 \leq i \leq L} \left\{ \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \right\} \leq 1$$

- ▶ The maximum form is used in Section II-B of the paper after defining the normalized matrix $\mathbf{X}_i$.
- ▶ It converts L separate constraints into one compact expression.
- ▶ If the largest normalized power value is less than or equal to 1, then all individual power constraints are satisfied.
- ▶ This is useful because the original optimization problem can now write the mixed power constraints as one single constraint.
- ▶ $\mathbf{S}_i$: original matrix selecting the antenna, antenna group, or total transmit power being constrained.
- ▶ L : total number of power constraints.

Layer 4: QoS Constraint

$$R_k(\rho_k(\mathbf{V})) \geq \widehat{R}_k, \quad \forall k = 1, \dots, K$$

$$\rho_k(\mathbf{V}) \geq \widehat{\rho}_k, \quad \forall k = 1, \dots, K$$

$$R_k(\widehat{\rho}_k) = \widehat{R}_k$$

- ▶ QoS means each user must receive at least a minimum required rate.
- ▶ $R_k(\rho_k(\mathbf{V}))$: finite-blocklength rate achieved by user k .
- ▶ $\widehat{R}_k$: minimum required rate for user k .
- ▶ $\rho_k(\mathbf{V})$: SINR achieved by user k using precoding matrix $\mathbf{V}$.
- ▶ $\widehat{\rho}_k$: minimum SINR needed to achieve rate $\widehat{R}_k$.
- ▶ Since $R_k(\rho_k)$ increases with ρ_k , the rate constraint can be rewritten as an SINR constraint.

Problems with this QoS Transformation

$$R_k(\rho_k) = \log(1 + \rho_k) - \frac{Q^{-1}(\epsilon_k)}{\sqrt{M}} \sqrt{B(\rho_k)}$$

$$B(\rho_k) = 1 - (1 + \rho_k)^{-2}$$

- ▶ The paper rewrites the QoS constraint:

$$R_k(\rho_k(\mathbf{V})) \geq \hat{R}_k$$

as:

$$\rho_k(\mathbf{V}) \geq \hat{\rho}_k$$

- ▶ This requires $R_k(\rho_k)$ to be increasing with respect to ρ_k .
- ▶ However, the dispersion term also increases with SINR:

$$\frac{dB(\rho_k)}{d\rho_k} = 2(1 + \rho_k)^{-3} > 0$$

- ▶ Therefore, increasing SINR increases both:
 - ▶ the useful term $\log(1 + \rho_k)$,
 - ▶ the penalty term $\sqrt{B(\rho_k)}$.
- ▶ The transformation is valid only when the useful rate increase dominates the penalty increase, ie: the shannon rate term increase dominates over channel dispersion term. on the valid increasing region of the FBL rate curve.

Layer 5: Original Optimization Problem

$$\begin{aligned} & \max_{\mathbf{V}} \quad \sum_{k=1}^K w_k R_k(\rho_k(\mathbf{V})) \\ & \text{subject to} \quad \max_{1 \leq i \leq L} \left\{ \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \right\} \leq 1 \\ & \quad \rho_k(\mathbf{V}) \geq \hat{\rho}_k, \quad \forall k = 1, \dots, K \end{aligned}$$

- ▶ $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_K]$: complete precoding matrix.
- ▶ w_k : priority weight of user k .
- ▶ $R_k(\rho_k(\mathbf{V}))$: FBL rate of user k .
- ▶ $\mathbf{X}_i$: normalized power constraint matrix.
- ▶ $\hat{\rho}_k$: minimum required SINR for user k .
- ▶ The problem is difficult because $\rho_k(\mathbf{V})$ is non-convex in $\mathbf{V}$.

Layer 6: Why Directly Solving the Problem is Hard

$$R_k(\rho_k) = \log(1 + \rho_k) - \frac{Q^{-1}(\epsilon_k)}{\sqrt{M}} \sqrt{1 - (1 + \rho_k)^{-2}}$$

$$\rho_k(\mathbf{V}) = \frac{|\mathbf{h}_k^H \mathbf{D}_k \mathbf{v}_k|^2}{\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{D}_j \mathbf{v}_j|^2 + \sigma_k^2}$$

- ▶ The difficulty comes from two sources.
- ▶ **First: SINR is non-convex in the precoders.**

$$\rho_k(\mathbf{V})$$

is a ratio of quadratic functions of the precoding vectors.

- ▶ Improving one user's signal power can increase interference to other users.
- ▶ **Second: the finite-blocklength rate is not the classical Shannon rate.**
- ▶ In the infinite-blocklength case, the rate is:

$$\log(1 + \rho_k)$$

- ▶ In the finite-blocklength case, the rate contains the extra dispersion penalty:

$$\frac{Q^{-1}(\epsilon_k)}{\sqrt{M}} \sqrt{1 - (1 + \rho_k)^{-2}}$$

- ▶ Because of this penalty term, standard infinite-blocklength precoding tools such as closed-form WMMSE updates are not directly applicable.
- ▶ Therefore, the paper avoids directly optimizing $\mathbf{V}$ and later reformulates the problem using target SINR allocation.

Layer 6: Main Reformulation Idea

$$\boldsymbol{\gamma} = [\gamma_1, \gamma_2, \dots, \gamma_K]^T$$

- ▶ Instead of directly optimizing the precoders $\mathbf{V}$, the paper optimizes target SINR values.
- ▶ γ_k : target SINR assigned to user k .
- ▶ $\boldsymbol{\gamma} \in \mathbb{R}^K$: vector containing the target SINRs of all users.
- ▶ The original problem asks:

What precoders $\mathbf{V}$ maximize the FBL weighted sum rate?

- ▶ The reformulated problem asks:

What SINR targets γ_k should users receive?

- ▶ After choosing $\boldsymbol{\gamma}$, the paper checks whether there exists a precoding matrix $\mathbf{V}$ that can achieve those SINR targets under the mixed power constraints.
- ▶ This separates the problem into:

SINR allocation + precoder feasibility checking.

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Layer 6: Theorem 1 – Equivalent SINR Allocation

Theorem 1

The original precoder optimization problem over $\mathbf{V}$ is equivalent to an optimization problem over target SINRs γ .

$$\max_{\mathbf{V}} \sum_{k=1}^K w_k R_k(\rho_k(\mathbf{V})) \iff \max_{\gamma} \sum_{k=1}^K w_k R_k(\gamma_k)$$

subject to $P(\gamma) \leq 1, \quad \gamma \geq \hat{\rho}$

- ▶ $\mathbf{V}$: original precoding matrix.
- ▶ $\rho_k(\mathbf{V})$: actual SINR achieved by user k .
- ▶ γ : target SINR vector chosen by the new problem.
- ▶ $P(\gamma)$: minimum normalized power needed to achieve γ .
- ▶ $P(\gamma) \leq 1$: the target SINRs are feasible under the mixed power constraints.
- ▶ Theorem 1 is the key step that changes the problem from designing precoders directly to allocating feasible SINR targets.

Layer 6: Definition of $P(\gamma)$

$$P(\gamma) = \min_{\mathbf{V}} \max_{1 \leq i \leq L} \left\{ \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \right\}$$

subject to $\rho_k(\mathbf{V}) \geq \gamma_k, \quad \forall k = 1, \dots, K$

- ▶ $P(\gamma)$: minimum normalized power required to achieve the target SINR vector γ .
- ▶ $\gamma = [\gamma_1, \dots, \gamma_K]^T$: target SINR vector.
- ▶ γ_k : target SINR required for user k .
- ▶ $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_K]$: precoding matrix being optimized.
- ▶ $\mathbf{X}_i$: normalized matrix for power constraint i .
- ▶ If $P(\gamma) \leq 1$, the SINR target is feasible.
- ▶ If $P(\gamma) > 1$, the SINR target requires more power than allowed.

Layer 6: Why $P(\gamma)$ is a Min–Max Problem

$$P(\gamma) = \min_{\mathbf{V}} \max_{1 \leq i \leq L} \left\{ \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \right\}$$

$$\text{subject to } \rho_k(\mathbf{V}) \geq \gamma_k, \quad \forall k = 1, \dots, K$$

- After choosing target SINRs

$$\gamma = [\gamma_1, \dots, \gamma_K]^T,$$

the paper must check whether some precoder $\mathbf{V}$ can achieve them.

- The constraint

$$\rho_k(\mathbf{V}) \geq \gamma_k$$

means user k 's achieved SINR must be at least its target SINR.

- The inner maximum appears because the system has L normalized mixed power constraints:

$$\sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \leq 1, \quad i = 1, \dots, L.$$

- Therefore,

$$\max_{1 \leq i \leq L} \left\{ \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \right\}$$

measures the worst violated or most heavily used normalized power constraint.

- The outer minimization over $\mathbf{V}$ searches for the best precoder that achieves γ using the least possible normalized power.

Layer 6: Why the Outer Minimization is Added

$$P(\gamma) = \min_{\mathbf{V}} \max_{1 \leq i \leq L} \left\{ \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \right\}$$

$$\text{subject to } \rho_k(\mathbf{V}) \geq \gamma_k, \quad \forall k$$

- ▶ After choosing a target SINR vector γ , many different precoders $\mathbf{V}$ may achieve it.
- ▶ The paper does not want to check only one arbitrary precoder.
- ▶ Instead, it asks:

Among all precoders that achieve γ , which one uses the least worst-case normalized power?

- ▶ This is why the outer minimization is added:

$$\min_{\mathbf{V}}$$

- ▶ The inner maximum measures the worst power constraint for a chosen $\mathbf{V}$.
- ▶ The outer minimum finds the best possible $\mathbf{V}$ that minimizes that worst constraint.
- ▶ Therefore:

$$P(\gamma) = \text{minimum worst-case normalized power needed to achieve } \gamma.$$

- ▶ If $P(\gamma) \leq 1$, then at least one feasible precoder exists.

Layer 7: Why Solving $P(\gamma)$ is Still Difficult

$$P(\gamma) = \min_{\mathbf{V}} \max_{1 \leq i \leq L} \left\{ \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \right\}$$

$$\text{subject to } \rho_k(\mathbf{V}) \geq \gamma_k, \quad \forall k = 1, \dots, K$$

- ▶ Even after the SINR-allocation reformulation, computing $P(\gamma)$ is not trivial.
- ▶ The objective contains a maximum over L mixed power constraints.
- ▶ The constraints still contain the SINR expression:

$$\rho_k(\mathbf{V}) = \frac{|\mathbf{h}_k^H \mathbf{D}_k \mathbf{v}_k|^2}{\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{D}_j \mathbf{v}_j|^2 + \sigma_k^2}.$$

- ▶ This SINR constraint couples all users because each $\mathbf{v}_j$ can create interference for user k .
- ▶ Therefore, the paper next transforms this power feasibility problem into an equivalent dual-domain convex problem.

Layer 7: Converting $P(\gamma)$ into a Dual Problem

$$P(\gamma) = \min_{\mathbf{V}} \max_{1 \leq i \leq L} \left\{ \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k \right\}$$

$$\text{subject to } \rho_k(\mathbf{V}) \geq \gamma_k, \quad \forall k = 1, \dots, K$$

- ▶ The paper does not solve this constrained minimization directly.
- ▶ Instead, it converts Problem (11) into an equivalent dual-domain problem.
- ▶ New variables are introduced:

$$\mathbf{q} = [q_1, \dots, q_L]^T, \quad \boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_K]^T$$

- ▶ q_i : dual variable associated with power constraint i .
- ▶ λ_k : dual variable associated with the SINR constraint

$$\rho_k(\mathbf{V}) \geq \gamma_k.$$

- ▶ The goal is to compute $P(\gamma)$ efficiently while accounting for both mixed power constraints and SINR target constraints.

Layer 7: Lemma 1 – Dual Form of $P(\gamma)$

$$\max_{\mathbf{q}, \boldsymbol{\lambda}} \quad \sum_{k=1}^K \lambda_k$$

subject to $\mathbf{A}_k(\mathbf{q}, \boldsymbol{\lambda}) \succeq 0, \quad \forall k = 1, \dots, K$

$$\sum_{i=1}^L q_i = 1, \quad \mathbf{q} \geq 0, \quad \boldsymbol{\lambda} \geq 0$$

- ▶ Lemma 1 states that the difficult power problem $P(\gamma)$ has the same optimal value as this dual problem.
- ▶ $\mathbf{q} = [q_1, \dots, q_L]^T$: dual variables for the L mixed power constraints.
- ▶ q_i : weight assigned to power constraint i .
- ▶ $\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_K]^T$: dual variables for the K SINR constraints.
- ▶ λ_k : dual variable associated with user k 's SINR target γ_k .
- ▶ $\mathbf{A}_k(\mathbf{q}, \boldsymbol{\lambda}) \succeq 0$: matrix feasibility condition for user k .
- ▶ $\succeq 0$: positive semidefinite matrix condition.

Layer 7: Meaning of $\mathbf{A}_k(\mathbf{q}, \lambda)$

$$\mathbf{A}_k(\mathbf{q}, \lambda) = \sum_{i=1}^L q_i \mathbf{X}_i + \mathbf{D}_k \left(\sum_{j \neq k} \lambda_j \mathbf{h}_j \mathbf{h}_j^H - \frac{\lambda_k}{\gamma_k} \mathbf{h}_k \mathbf{h}_k^H \right) \mathbf{D}_k$$

$$\mathbf{A}_k(\mathbf{q}, \lambda) \succeq 0, \quad \forall k$$

- ▶ $\mathbf{A}_k(\mathbf{q}, \lambda)$: matrix condition for user k in the dual problem.
- ▶ $\sum_{i=1}^L q_i \mathbf{X}_i$: weighted combination of all normalized power constraints.
- ▶ $\lambda_j \mathbf{h}_j \mathbf{h}_j^H$: interference-related term from user j .
- ▶ $\frac{\lambda_k}{\gamma_k} \mathbf{h}_k \mathbf{h}_k^H$: desired-signal term for user k , scaled by its SINR target.
- ▶ $\mathbf{D}_k$: selects antennas that serve user k .
- ▶ $\mathbf{A}_k \succeq 0$ ensures the dual function remains bounded.

Layer 7: Where $\mathbf{A}_k$ Comes From

$$p_i(\mathbf{V}) = \sum_{k=1}^K \mathbf{v}_k^H \mathbf{X}_i \mathbf{v}_k$$

$$\max_i p_i(\mathbf{V}) \implies p_i(\mathbf{V}) \leq \xi, \quad \forall i$$

$$\mathcal{L}(\xi, \mathbf{V}, \mathbf{q}, \lambda) = \xi + \sum_{i=1}^L q_i(p_i(\mathbf{V}) - \xi) - \sum_{k=1}^K \lambda_k \left(\frac{1}{\gamma_k} |\mathbf{h}_k^H \mathbf{D}_k \mathbf{v}_k|^2 - \sum_{j \neq k} |\mathbf{h}_k^H \mathbf{D}_j \mathbf{v}_j|^2 - 1 \right)$$

- ▶ ξ : slack variable replacing the maximum power term.
- ▶ $p_i(\mathbf{V})$: normalized power used under constraint i .
- ▶ q_i : dual variable for $p_i(\mathbf{V}) \leq \xi$.
- ▶ λ_k : dual variable for user k 's SINR constraint.
- ▶ After collecting all terms containing each $\mathbf{v}_k$, the quadratic coefficient becomes $\mathbf{A}_k(\mathbf{q}, \lambda)$.

Layer 7: From Lagrangian to the Dual Problem

$$\mathcal{L} = \sum_{k=1}^K \lambda_k - \xi \left(\sum_{i=1}^L q_i - 1 \right) + \sum_{k=1}^K \mathbf{v}_k^H \mathbf{A}_k(\mathbf{q}, \lambda) \mathbf{v}_k$$

$$\sum_{i=1}^L q_i = 1, \quad \mathbf{A}_k(\mathbf{q}, \lambda) \succeq 0, \quad \forall k$$

$$\Rightarrow \max_{\mathbf{q}, \lambda} \sum_{k=1}^K \lambda_k$$

$$\text{subject to } \mathbf{A}_k(\mathbf{q}, \lambda) \succeq 0, \quad \forall k, \quad \sum_{i=1}^L q_i = 1, \quad \mathbf{q} \geq 0, \quad \lambda \geq 0$$

- ▶ The dual problem keeps only values of $\mathbf{q}$ and λ that make the Lagrangian bounded.
- ▶ $\sum_i q_i = 1$: removes the unbounded slack-variable term involving ξ .
- ▶ $\mathbf{A}_k \succeq 0$: prevents the quadratic precoder term from going to $-\infty$.
- ▶ After these conditions hold, minimizing over ξ and $\mathbf{V}$ leaves the dual objective $\sum_k \lambda_k$.

Layer 8: Solving the Dual Problem – Updating λ

$$\lambda_k \leftarrow \frac{\gamma_k}{\mathbf{h}_k^H \mathbf{D}_k \mathbf{C}_k^{-1} \mathbf{D}_k \mathbf{h}_k}, \quad \forall k = 1, \dots, K$$

$$\mathbf{C}_k = \mathbf{D}_k \left(\sum_{j \neq k} \lambda_j \mathbf{h}_j \mathbf{h}_j^H \right) \mathbf{D}_k + \sum_{i=1}^L q_i \mathbf{X}_i$$

- ▶ This update is used inside Algorithm 1 to solve the dual problem from Lemma 1.
- ▶ λ_k : dual variable associated with user k 's SINR target constraint.
- ▶ γ_k : target SINR assigned to user k .
- ▶ $\mathbf{C}_k$: effective interference-plus-power matrix seen while updating user k 's dual variable.
- ▶ $\sum_{j \neq k} \lambda_j \mathbf{h}_j \mathbf{h}_j^H$: weighted interference contribution from all other users.
- ▶ $\sum_{i=1}^L q_i \mathbf{X}_i$: weighted mixed-power-constraint contribution.
- ▶ The denominator measures how strongly user k 's channel can support its SINR target under the current interference and power weights.
- ▶ The update increases λ_k when achieving γ_k is harder.

Layer 8: Reducing the Cost of Updating λ

$$\lambda_k \leftarrow \frac{\gamma_k}{\mathbf{h}_k^H \mathbf{D}_k \mathbf{C}_k^{-1} \mathbf{D}_k \mathbf{h}_k}$$

$$\mathbf{C}_k = \mathbf{D}_k \left(\sum_{j \neq k} \lambda_j \mathbf{h}_j \mathbf{h}_j^H \right) \mathbf{D}_k + \sum_{i=1}^L q_i \mathbf{X}_i$$

- ▶ Directly computing $\mathbf{C}_k^{-1}$ for every user is expensive.
- ▶ A full matrix inverse has complexity:

$$\mathcal{O}(N^3)$$

- ▶ Since the update is needed for all K users, this can become computationally costly.
- ▶ The paper avoids repeated full inversions by updating the inverse recursively using the matrix inversion lemma.
- ▶ This reduces the complexity to approximately:

$$\mathcal{O}(KN^2)$$

- ▶ N : number of transmit antennas.
- ▶ K : number of users.
- ▶ $\mathbf{C}_k^{-1}$: inverse matrix needed to update λ_k .

Layer 8: Matrix Inversion Lemma Update

$$\mathbf{C}_{k+1} = \mathbf{C}_k - \lambda_{k+1} \mathbf{D}_{k+1} \mathbf{h}_{k+1} \mathbf{h}_{k+1}^H \mathbf{D}_{k+1} + \lambda_k \mathbf{D}_k \mathbf{h}_k \mathbf{h}_k^H \mathbf{D}_k$$

$$\mathbf{C}_{k+1}^{-1} = \mathbf{C}_k^{-1} - \mathbf{C}_k^{-1} \mathbf{U}_k \mathbf{G}_k^{-1} \mathbf{W}_k \mathbf{C}_k^{-1}$$

- ▶ The paper avoids recomputing $\mathbf{C}_{k+1}^{-1}$ from scratch.
- ▶ $\mathbf{C}_k$: matrix used when updating λ_k .
- ▶ $\mathbf{C}_{k+1}$: next matrix used when updating λ_{k+1} .
- ▶ Since $\mathbf{C}_{k+1}$ differs from $\mathbf{C}_k$ by low-rank terms, its inverse can be updated efficiently.
- ▶ $\mathbf{U}_k$, $\mathbf{W}_k$, and $\mathbf{G}_k$: auxiliary matrices used in the matrix inversion lemma.
- ▶ This reduces repeated inverse computation and makes Algorithm 1 computationally efficient.

Layer 8: Updating $\mathbf{q}$ with Projected Subgradient Ascent

$$\tilde{\mathbf{q}}^{(n)} = \mathbf{q}^{(n)} + \eta^{(n)} (\mathbf{q}^{(n)} - \mathbf{q}^{(n-1)})$$

$$\mathbf{q}^{(n+1)} = P_{\mathcal{Q}} (\tilde{\mathbf{q}}^{(n)} + b^{(n)} \mathbf{g}(\tilde{\mathbf{q}}^{(n)}))$$

$$\mathcal{Q} = \left\{ \mathbf{q} \mid \sum_{i=1}^L q_i = 1, \mathbf{q} \geq 0 \right\}$$

$$P_{\mathcal{Q}}(\mathbf{a}) = \arg \min_{\mathbf{q} \in \mathcal{Q}} \|\mathbf{q} - \mathbf{a}\|_2^2$$

- ▶ $\mathbf{q}$: dual weight vector for the mixed power constraints.
- ▶ q_i : weight assigned to power constraint i .
- ▶ $\tilde{\mathbf{q}}^{(n)}$: momentum-based extrapolated value of $\mathbf{q}$.
- ▶ $\eta^{(n)}$: momentum coefficient that uses the previous movement direction.
- ▶ $b^{(n)}$: subgradient step size.
- ▶ $\mathbf{g}(\tilde{\mathbf{q}}^{(n)})$: subgradient showing which power constraints are heavily used.
- ▶ $\mathcal{Q}$: feasible set requiring $q_i \geq 0$ and $\sum_i q_i = 1$.
- ▶ $P_{\mathcal{Q}}(\cdot)$: moves the updated vector back to the closest valid $\mathbf{q}$.
- ▶ This is needed because the subgradient step may produce negative entries or make $\sum_i q_i \neq 1$.

Layer 8: Subgradient Used to Update $\mathbf{q}$

$$g_i(\tilde{\mathbf{q}}^{(n)}) = \sum_{k=1}^K \left(\mathbf{v}_k^{(n)} \right)^H \mathbf{X}_i \mathbf{v}_k^{(n)}$$

$$\mathbf{v}_k = \sqrt{\delta_k} \mathbf{C}_k^{-1} \mathbf{D}_k \mathbf{h}_k$$

- ▶ $g_i(\tilde{\mathbf{q}}^{(n)})$: subgradient component for power constraint i .
- ▶ It measures how much normalized power is used under constraint i .
- ▶ $\mathbf{v}_k^{(n)}$: precoder for user k at inner iteration n .
- ▶ $\mathbf{X}_i$: normalized matrix for power constraint i .
- ▶ δ_k : power coefficient used to construct the precoder.
- ▶ $\mathbf{C}_k^{-1} \mathbf{D}_k \mathbf{h}_k$: beamforming direction for user k .
- ▶ The update increases q_i when constraint i is heavily used.

Layer 8: Algorithm 1 – Computing $P(\gamma)$

1. Initialize $\mathbf{q}^{(0)}$.
2. Compute the extrapolated point:

$$\tilde{\mathbf{q}}^{(n)} = \mathbf{q}^{(n)} + \eta^{(n)} (\mathbf{q}^{(n)} - \mathbf{q}^{(n-1)})$$

3. Update λ using the fixed-point rule:

$$\lambda_k \leftarrow \frac{\gamma_k}{\mathbf{h}_k^H \mathbf{D}_k \mathbf{C}_k^{-1} \mathbf{D}_k \mathbf{h}_k}$$

4. Construct the precoders:

$$\mathbf{v}_k = \sqrt{\delta_k} \mathbf{C}_k^{-1} \mathbf{D}_k \mathbf{h}_k$$

5. Update $\mathbf{q}$ by projected subgradient ascent.

- Output: the value $P(\gamma)$, the dual variables $\mathbf{q}^*$, λ^* , and the corresponding precoders $\mathbf{v}_k^*$.
- This algorithm is used every time the outer algorithm needs to check whether a target SINR vector γ is feasible.

Layer 10: BDGM – Main Idea

$$\mathbf{z} = \log(1 + \gamma)$$

$$z_k = \log(1 + \gamma_k), \quad k = 1, \dots, K$$

- ▶ After Theorem 1, the paper must solve the SINR-allocation problem:

$$\max_{\gamma} \sum_{k=1}^K w_k R_k(\gamma_k)$$

subject to:

$$P(\gamma) \leq 1, \quad \gamma \geq \hat{\rho}.$$

- ▶ BDGM stands for **Bi-Direction Gradient Method**.
- ▶ Instead of updating γ directly, BDGM updates:

$$\mathbf{z} = \log(1 + \gamma).$$

- ▶ $\mathbf{z} = [z_1, \dots, z_K]^T$: rate-domain SINR variable.
- ▶ If the current point is feasible, BDGM moves forward to increase the weighted sum rate.
- ▶ If the current point is infeasible, BDGM moves backward to reduce the required power.

Layer 10: BDGM Forward Step

$$P_z(\mathbf{z}^{(t)}) \leq 1$$

$$\mathbf{z}^{(t+1)} = \mathbf{z}^{(t)} + \alpha^{(t)} \nabla f_z(\mathbf{z}^{(t)})$$

$$[\nabla f_z(\mathbf{z})]_k = w_k \left(1 - \frac{Q^{-1}(\epsilon_k)}{\sqrt{M}} \frac{e^{-2z_k}}{\sqrt{1 - e^{-2z_k}}} \right)$$

- ▶ This step is used when the current SINR target is feasible.
- ▶ t : outer iteration index.
- ▶ $\mathbf{z}^{(t)}$: current rate-domain SINR vector.
- ▶ $P_z(\mathbf{z}) = P(e^{\mathbf{z}} - \mathbf{1})$: required normalized power in the $\mathbf{z}$ -domain.
- ▶ $f_z(\mathbf{z}) = f(e^{\mathbf{z}} - \mathbf{1})$: weighted sum-rate objective in the $\mathbf{z}$ -domain.
- ▶ $\alpha^{(t)}$: forward step size.
- ▶ $\nabla f_z(\mathbf{z}^{(t)})$: direction that increases weighted sum rate.
- ▶ Intuition: if the current point is feasible, BDGM increases the users' SINR targets to improve rate.

Layer 10: BDGM Backward Step

$$P_z(\mathbf{z}^{(t)}) > 1$$

$$\mathbf{z}^{(t+1)} = \max \left(\mathbf{z}^{(t)} - \beta^{(t)} \nabla P_z(\mathbf{z}^{(t)}), \log(1 + \hat{\rho}) \right)$$

$$[\nabla P_z(\mathbf{z})]_k = e^{z_k} \frac{\partial P(\gamma)}{\partial \gamma_k} \bigg|_{\gamma = e^{\mathbf{z}} - 1}$$

- ▶ This step is used when the current SINR target is infeasible.
- ▶ $P_z(\mathbf{z}^{(t)}) > 1$: current target SINRs require more normalized power than allowed.
- ▶ $\beta^{(t)}$: backward step size.
- ▶ $\nabla P_z(\mathbf{z}^{(t)})$: direction in which required power increases fastest.
- ▶ Moving in the negative gradient direction reduces the required power.
- ▶ $\log(1 + \hat{\rho})$: lower bound that prevents violating QoS.
- ▶ The max operation ensures the updated SINR target stays above the QoS threshold.

Layer 10: BDGM Overall Procedure

$$\mathbf{z} = \log(1 + \gamma)$$

1. Start with an initial SINR target vector:

$$\gamma^{(0)}$$

2. Convert it to the rate-domain variable:

$$\mathbf{z}^{(0)} = \log(1 + \gamma^{(0)})$$

3. At iteration t , compute:

$$P_z(\mathbf{z}^{(t)})$$

4. If:

$$P_z(\mathbf{z}^{(t)}) \leq 1,$$

use the forward step to increase the objective.

5. If:

$$P_z(\mathbf{z}^{(t)}) > 1,$$

use the backward step to recover feasibility.

- ▶ BDGM alternates between rate improvement and power-feasibility correction.
- ▶ The output is a SINR target vector that satisfies QoS and mixed power constraints.

Layer 10: BDGM Convergence

$P_z(\mathbf{z}) \leq 1$ means feasible

$P_z(\mathbf{z}) > 1$ means infeasible

$$\nabla f_z(\mathbf{z}^*) = \mu \nabla P_z(\mathbf{z}^*), \quad \mu \geq 0$$

- ▶ The paper proves that BDGM converges to a point satisfying the KKT conditions of the SINR-allocation problem.
- ▶ $\mathbf{z}^*$: limiting point reached by BDGM.
- ▶ $\nabla f_z(\mathbf{z}^*)$: gradient of the weighted sum-rate objective.
- ▶ $\nabla P_z(\mathbf{z}^*)$: gradient of the required normalized power function.
- ▶ μ : non-negative multiplier for the feasibility boundary $P_z(\mathbf{z}) \leq 1$.
- ▶ At convergence, the forward rate-improvement direction and backward feasibility-correction direction balance each other.
- ▶ This means BDGM stops at a stationary feasible SINR allocation.

Layer 11: Local Linearization of $P(\gamma) \leq 1$

$$P(\gamma^{(t)}) + \nabla P(\gamma^{(t)})^T (\gamma - \gamma^{(t)}) \leq 1$$

- ▶ IWF approximates the difficult constraint $P(\gamma) \leq 1$ around the current point $\gamma^{(t)}$.
- ▶ $\gamma^{(t)}$: SINR target vector at iteration t .
- ▶ γ : candidate SINR target vector for the next update.
- ▶ $P(\gamma^{(t)})$: required normalized power at the current SINR target.
- ▶ $\nabla P(\gamma^{(t)})$: gradient of required power with respect to SINR targets.
- ▶ The linearized constraint estimates how the required power changes when moving from $\gamma^{(t)}$ to γ .
- ▶ This turns the difficult feasibility constraint into a simpler linear constraint for the next update.

Layer 11: IWF Water-Filling Optimality Condition

$$c_k^{(t)} = [\nabla P(\gamma^{(t)})]_k$$

$$w_k \frac{\partial R_k(\hat{\gamma}_k^{(t+1)})}{\partial \gamma_k} = \nu c_k^{(t)}, \quad \forall k$$

$$\mathbf{g}^{(t)} = \mathbf{1} - P(\gamma^{(t)}) + \nabla P(\gamma^{(t)})^T \gamma^{(t)}$$

$$\nabla P(\gamma^{(t)})^T \hat{\gamma}^{(t+1)} = \mathbf{g}^{(t)}$$

- ▶ $\mathbf{g}^{(t)}$: available linearized power budget at iteration t .
- ▶ $c_k^{(t)}$: marginal power cost of increasing user k 's SINR target at iteration t .
- ▶ $\nu \geq 0$: dual variable, interpreted as the water level.
- ▶ w_k : priority weight of user k .
- ▶ $\frac{\partial R_k}{\partial \gamma_k}$: marginal FBL rate gain from increasing γ_k .
- ▶ The condition balances marginal rate gain against marginal power cost.
- ▶ ν is found by bisection so that the linearized power constraint is satisfied.

Layer 11: Why Gradients Balance at the IWF Optimum

$$\max_{\gamma \geq \hat{\rho}} f(\gamma) \quad \text{subject to} \quad \sum_{k=1}^K c_k^{(t)} \gamma_k \leq g^{(t)}$$

$$c_k^{(t)} = [\nabla P(\gamma^{(t)})]_k$$

$$\mathcal{L} = f(\gamma) - \nu \left(\sum_{k=1}^K c_k^{(t)} \gamma_k - g^{(t)} \right)$$

$$\frac{\partial \mathcal{L}}{\partial \gamma_k} = 0$$

$$w_k \frac{\partial R_k(\gamma_k)}{\partial \gamma_k} = \nu c_k^{(t)}$$

- ▶ $f(\gamma) = \sum_{k=1}^K w_k R_k(\gamma_k)$: weighted FBL sum-rate.
- ▶ The equality comes from setting the derivative of the Lagrangian to zero.
- ▶ It means marginal weighted rate gain equals marginal power cost multiplied by the common price ν .

Layer 11: Smoothed IWF Update

$$\gamma^{(t+1)} = \eta^{(t)} \hat{\gamma}^{(t+1)} + (1 - \eta^{(t)}) \gamma^{(t)}$$

- ▶ $\hat{\gamma}^{(t+1)}$: candidate SINR vector obtained from the water-filling subproblem.
- ▶ $\gamma^{(t)}$: current SINR target vector.
- ▶ $\gamma^{(t+1)}$: actual next SINR target vector used by the algorithm.
- ▶ $\eta^{(t)} \in (0, 1]$: smoothing parameter or step size.
- ▶ If $\eta^{(t)} = 1$, the algorithm fully moves to the water-filling solution.
- ▶ If $\eta^{(t)} < 1$, the algorithm moves only partially toward the candidate solution.
- ▶ This smoothing improves stability and helps maintain feasibility.

Layer 11: IWF Overall Algorithm

1. Initialize the SINR target vector:

$$\gamma^{(0)}$$

2. At iteration t , compute:

$$P(\gamma^{(t)}) \quad \text{and} \quad \nabla P(\gamma^{(t)})$$

3. Locally linearize the power feasibility constraint:

$$P(\gamma) \leq 1$$

4. Solve the water-filling subproblem to obtain:

$$\hat{\gamma}^{(t+1)}$$

5. Apply the smoothed update:

$$\gamma^{(t+1)} = \eta^{(t)} \hat{\gamma}^{(t+1)} + (1 - \eta^{(t)}) \gamma^{(t)}$$

- ▶ $P(\gamma^{(t)})$: required normalized power at the current SINR target.
- ▶ $\nabla P(\gamma^{(t)})$: marginal power cost of increasing SINR targets.
- ▶ $\hat{\gamma}^{(t+1)}$: candidate water-filling SINR vector.
- ▶ $\eta^{(t)}$: smoothing parameter.